CMSE 202 Project Final Report

Group members: Omar Abdulwahhab, Mohammed Alsayegh, Mint Menakanit, Prisha Kuma, Sai Palaka

Introduction

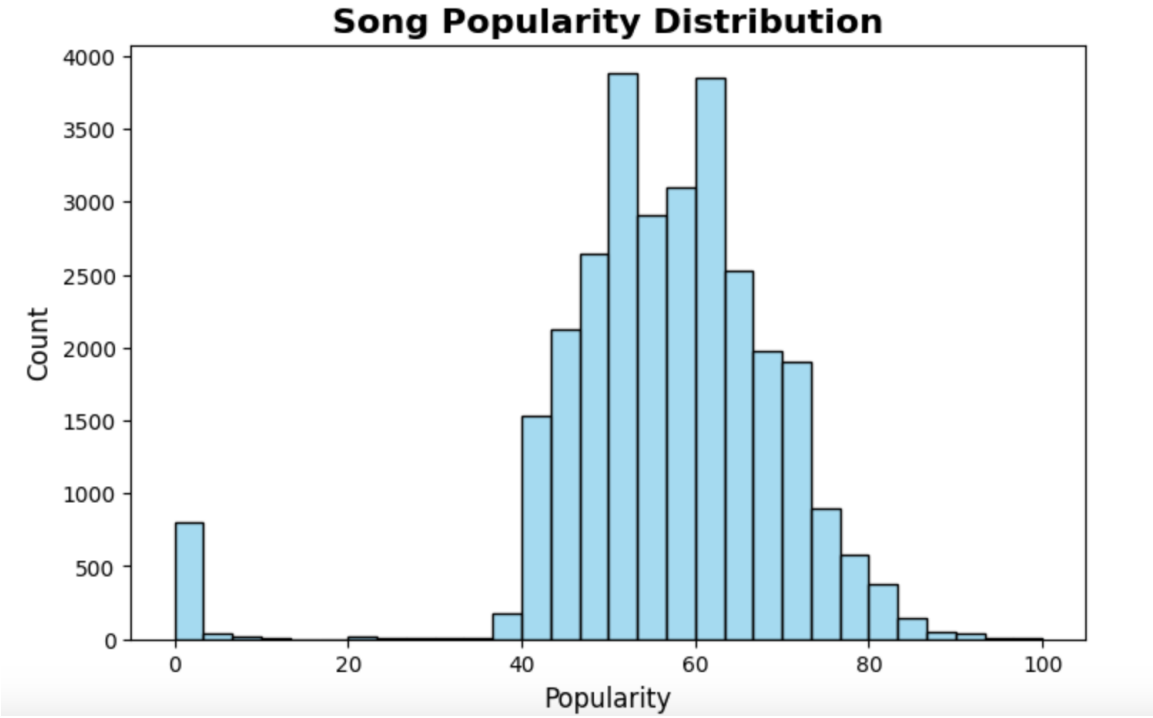
In this project, we aimed to answer a simple yet interesting question: what makes a song popular on Spotify? With millions of users and a huge catalog of music, Spotify tracks many audio features and assigns each song a popularity score between 0 and 100. We wanted to know which features—danceability, energy, tempo, or loudness, for example—go into making a song more popular.

Methodology

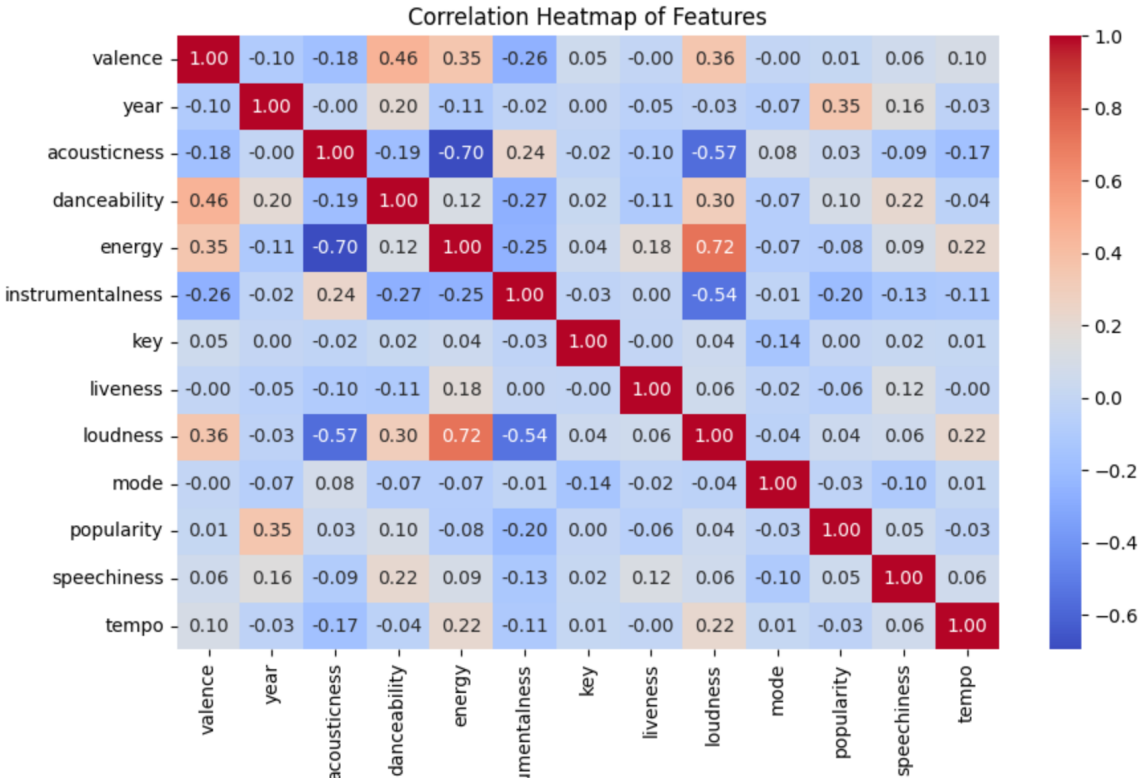
We used a Kaggle dataset that included over 160,000 tracks from the year 2006 to 2020, with each track including extensive information about various music-related aspects. We first imported required Python libraries like Pandas, Seaborn, Matplotlib, and Scikit-learn, then filtered the data to include songs from the year 2006 and beyond. We inspected the data structure and cleaned the data by removing any missing values so that we could ensure our analysis would be valid.

Data Analysis

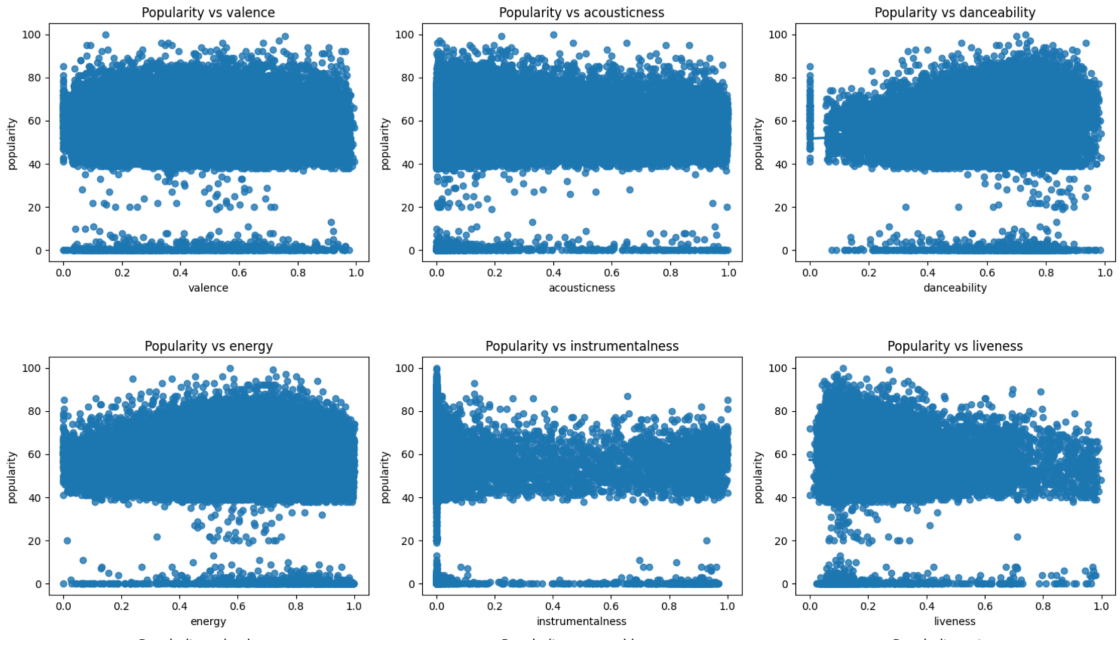
The distribution of song popularity scores:

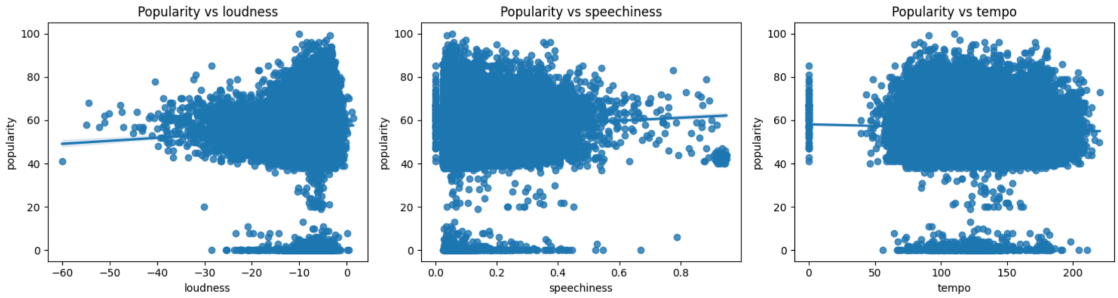


Correlation heatmap showing relationships between features and popularity:



scatter plot comparing each features to popularity to find more in depth relationships





Results

We observed that more recently released songs were more popular, and features like energy and loudness were also positively correlated moderately. Acousticness had a reverse relation with popularity, i.e., more acoustic songs were less popular on average. Scatter plots showed that more energetic and happier songs were more likely to fall on the higher side of the popularity scores.

Conclusion

While external variables like artist popularity or social media following also play a role in the success of a song, musical features—especially energy, loudness, and the newness of a song—do have a measurable effect. In the future, we could expand the model to include other features like genre, artist popularity, or playlist position to allow predictions to be even more accurate.